

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE 1ST SEMESTER SECOND INTERNAL EXAMINATIONS, DEC 2023
23-200-104B INTRODUCTION TO ELECTRICAL ENGINEERING
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO2	Develop & solve models of basic magnetic & electromagnetic circuits.
CO3	Apply elementary principles for finding the sinusoidal steady state features of circuits.

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q.No	Questions	Marks	BL	CO	PO
I (a)	Define form factor.	2	L2	2	1
(b)	Derive the expression for the coefficient of coupling in a magnetic circuit.	2	L2	2	1
(c)	Explain principle of operation of the transformer.	2	L2	2	1
(d)	An ac circuit consists of a pure resistance of 10 ohms and is connected across an ac supply of 230V, 50Hz. Write down the equation for instantaneous value of current.	2	L3	3	1
(e)	Represent trigonometric, polar, exponential and rectangular forms of phasor.	2	L1	3	1

PART B

(Answer the following questions
choosing either the first OR second option)

(1 x 10= 10 Marks)

Q.No	Questions	Marks	BL	CO	PO
II (a)	Derive the rms value for a sine wave.	5	L2	3	1
(b)	A coil consists of 750 turns and a current of 10 A in the coil produces a magnetic flux of 1.2 m Wb. Calculate the	5	L3	3	1

	inductance of the coil. If the current in the coil is reversed in 0.01 second, find the average emf induced in the coil.				
OR					
III (a)	An iron core in the form of a rectangular shape is arranged as a magnetic circuit. The core is wound with a coil of <u>1000</u> turns and cross-sectional area of the core is <u>10 cm²</u> . The mean length of the flux path through the iron core is <u>40 cm</u> and that of air gap is <u>0.01 cm</u> . Calculate the current required to produce a flux of 2 mWb in the core. Take the relative permeability of iron as 2000.	10	L3	2	1
(1 x 10= 10 Marks)					
IV (a)	Two coils A and B of turns 400 and 600 and of length 0.2 m and 0.3 m respectively are wound side by side on a same iron core of cross-sectional area 80 cm ² and relative permeability 800. Calculate self-inductance of the coil and mutual inductance if K=0.4.	10	L3	3	1
OR					
V (a)	Define the following terms in AC circuits, a. Time period b. Amplitude c. Phase d. Frequency e. Average Value	5	L1	3	1
(b)	Two currents i_1 and i_2 are given by the expressions $i_1 = 10\sin(\omega t + \pi/4)$ and $i_2 = 8\sin(\omega t + \pi/3)$. Find $i_1 + i_2$. Express the answer in the form $i = I_m \sin(\omega t + \phi)$?	5	L3	3	1

Bloom's Taxonomy Levels

L1 - %	L2- %	L3-%	L4-%	L5-%	L6-%
18.2	36.36	45.45			